CP 221 – JavaScript Practical Question

Question 1

A university wants to build an online student registration page. You have been tasked with writing a JavaScript function called validateForm() that:

- Prompts the user to enter their **full name**, **student ID** (must be a number), and **age** (must be between 18 and 35)
- Uses isNaN() to check that the student ID is numeric
- Uses appropriate comparison operators to validate the age range
- Displays an alert() if any field is invalid, or writes a welcome message to the document if all inputs are valid

Write the complete HTML page embedding this JavaScript in the <head> section.

Question 2

A local shop owner wants a simple web-based price calculator. Write a JavaScript program that:

- Ask the user to enter the **names and prices** of 3 items, storing prices in an **array**
- Calculates the **total**, **maximum price** and **average** of the three items
- Displays all results
- Handles the case where the user enters a non-numeric price using isNaN() and throws a custom exception with the message "InvalidPriceException"

Question 3

You are hired to build an online quiz for a school. Write a JavaScript program that:

- Stores **5 quiz questions** and their correct answers using **arrays**
- Uses a for loop to present each question via window.prompt()
- Uses a switch statement to give feedback: "Excellent!" for 5 correct, "Good" for 3–4, "Keep Practicing" for below 3
- Tracks the score using a variable and displays the **final score and grade** using document.writeln()
- Defines the grading logic inside a **function** called gradeStudent(score)

Question 4

A class teacher needs a program to manage student records. Write a JavaScript program that:

- Creates a **student object** using a constructor function Student(name, regNo, marks) with three properties
- Creates an **array of 4 student objects** using object literals or new
- Uses a for...in loop to display all properties of each student
- Includes a function topStudent(students) that iterates through the array and returns the student with the highest marks
- Displays the top student's name and marks using document.writeln()

Question 5

A developer is building a unit converter web page. Write a complete HTML page with embedded JavaScript that:

- Uses the navigator object to display the user's **browser name and platform** at the top of the page
- Prompts the user to enter a **temperature in Celsius** and choose a conversion: 1 for Fahrenheit, 2 for Kelvin
- Uses an if-else or switch statement to perform the correct conversion using the appropriate formula
- Wraps the conversion logic inside a **try-catch-finally** block — throwing a custom exception "InvalidChoiceException" if the user enters anything other than 1 or 2
- Displays the converted result and a confirm() box asking if the user wants to convert another value

Question 6

Use a one dimension array in Javascript to solve the following problem: Read in 20 numbers, each of which is between 10 and 100. As each number is read, print it only if it is not a duplicate of a number that has already been read. Provide for the “worst case”, in which all 20 numbers are different. Use the smallest possible array to solve this problem.

Question 7

Write a script that plays a “guess the number” game as follows: Your program chooses the number to be guessed by selecting a random integer in the range 1 to 100. The script displays the prompt Guess a number between 1 and 100 next to a field. The player types a first guess into the text field and clicks a button to submit the guess to the script. If the player’s guess is incorrect, your program should display Too High. Try again. or Too low. Try again. When the user enters the correct answer, display Congratulation. You guessed the number! And clear the text field so the user can play again. [Note The guessing technique employed in this problem is similar to a binary search.]

Question 8:

You are building a day-of-week greeting page. The script prompts the user for a number from 1 to 7 and should display a custom message for each day, for example, "Start of the week!" for Monday and "It's the weekend!" for Saturday and Sunday. A classmate implements this using seven separate if statements. You suggest using a switch statement instead. Using the control statement examples from the lecture, explain why switch is more appropriate here, write out the structure of the switch block for at least three of the days, and explain what would happen if a break statement were accidentally left out of one of the cases.

Question 9:

You are building a simple division calculator. The script prompts the user for two numbers, divides the first by the second, and displays the result. During testing, a classmate enters 0 as the second number, and the page displays Infinity. Another classmate enters the word "hello" and the result shows NaN. Using the isNaN() and isFinite() global functions, and the try/catch exception handling syntax, rewrite the script so that it detects and handles both of these invalid inputs gracefully — showing a user-friendly alert() message in each case instead of a confusing result.

Question 10:

You are building a discount pricing module for an e-commerce platform. A junior developer on your team writes a for loop using var to attach a click handler to each product card, expecting each handler to log a different price. During testing, every single card logs the same “the last price”. The junior developer is confused because the loop looks correct. How would you explain to them exactly what went wrong, and demonstrate at least two different ways to fix the problem without changing the fundamental structure of the feature?

Question 11:

The University of Dodoma has asked you to create a program that allows prospective students to provide feedback about their campus visit. Your program should contain a form with text boxes for a name, address and email. Provide checkboxes that allow prospective students to indicate what they liked most about the campus. The checkboxes should include: student, location, campus, atmosphere, dorm rooms and sports. Also, provide radio buttons that ask the prospective student how they became interested in the university. Options should include: friends, television, the Internet and others. In addition, provide a text area for additional comments and a JavaScript function to validate each form field. When a student clicks the submit button, the program should insert those form fields' values in this table tbl_feedback (name, address, email, yourLike, yourInterest, comments).